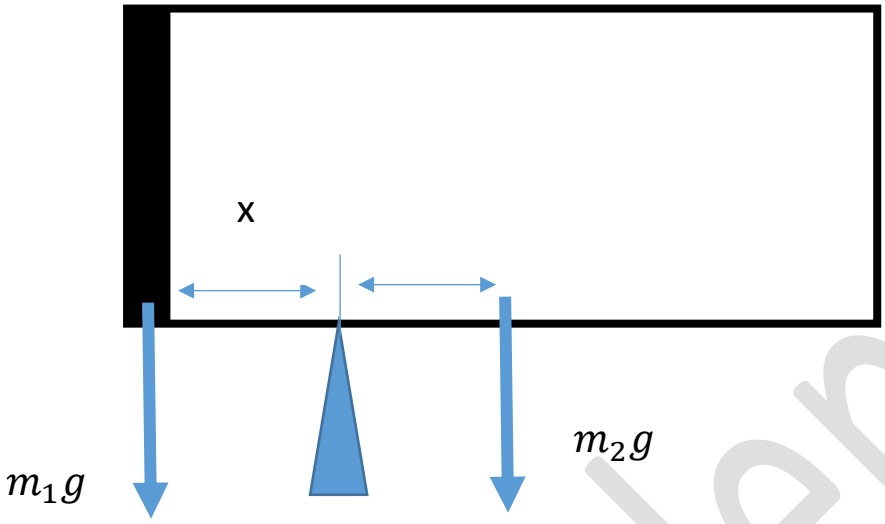
1- MASALA

Berilgan

H

D



y stakan devoir qalinligi u D dan juda kichik

$$m_1 g x = m_2 g(\frac{H}{2} - x)$$

$$m_1 = \rho * \pi * \frac{D^2}{4} * 3y$$

$$m_2 = \rho * \pi * D * y * H$$

$$\rho * \pi * \frac{D^2}{4} * 3y * x = \rho * \pi * D * y * H(\frac{H}{2} - x)$$

Ikkala tomondan ham y lar va ρ larni qisqartirsak

$$3Dx = 4 * H(\frac{H}{2} - x)$$

$$3Dx = 2H^2 - 4Hx$$

$$3Dx + 4Hx = 2H^2$$

$$x = \frac{2H^2}{3D + 4H}$$

$$javob: \quad x = \frac{2H^2}{3D + 4H}$$

@olimpdep

2- Masala

Nyutonning ikkinchi qonuniga ko’ra

$$5ma = 5mg + F$$

F=3mg eknaligini hisobga olsak

$$a = 1{,}6g$$

Ekanligi kelib chiqadi.

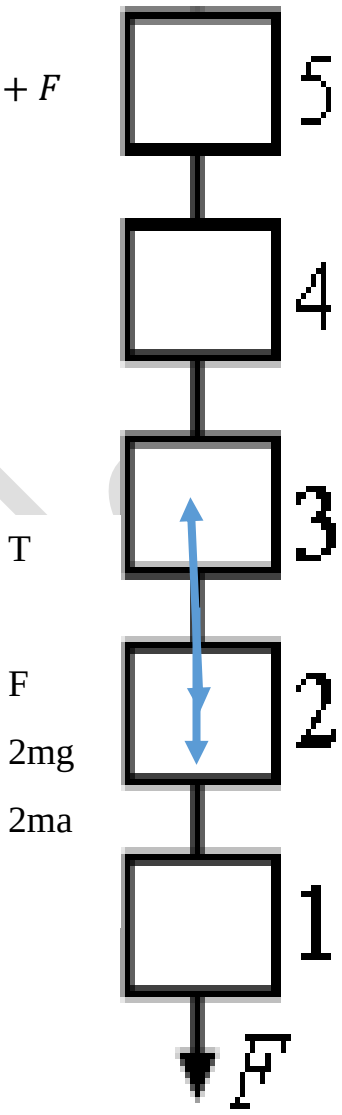
Ikkinchi jism uchun proyeksiya olsak

2ma=3mg+2mg-T

T=5mg-2ma

T=5mg-3,2mg= 1,8mg

Javob: T=1,8mg



@olimpdep

Fan olimpiadalari bo‘yicha
iqtidorli o‘quvchilar bilan ishlash
DEPARTAMENTI

3- Masala

Berilgan

d_1

h_1

d_2

h_2

F =?

$$\frac{1}{F}=\frac{1}{d_1}+\frac{1}{f_1}$$

$$\frac{1}{F}=\frac{1}{d_2}+\frac{1}{f_2}$$

$$\frac{f_1}{d_1}=\frac{h_1}{h}$$

$$\frac{f_2}{d_2}=\frac{h_2}{h}$$

$$f_1=\frac{h_1}{h}*d_1$$

$$h=\frac{d_2}{f_2}*h_2$$

$oxirgi\ ikki\ tenglamani\ 1-tenglamaga\ qo'ysak$

$\frac{1}{F}=\frac{1}{d_1}+\frac{1}{\frac{h_1*d_1}{h}}$ bundan $F=\frac{h_1+h}{h_1*d_1}$

Xuddi shu ishni ikkinchi tenglama uchun ham bajarsak

$$F=\frac{h_2+h}{h_2*d_2}$$

Bu ikkala tenglamani tenglashtirsak

$\frac{h_2+h}{h_2*d_2}=\frac{h_1+h}{h_1*d_1}$ bundan

$$(h_2+h)*h_1*d_1=(h_1+h)*h_2*d_2$$

$$h_1*d_1*h_2+h*h_1*d_1=h*h_2*d_2+h_1*h_2*d_2$$

$$h_1*h_2(d_1-d_2)=h(h_2*d_2-h_1*d_1)$$

$$h=\frac{h_1*h_2(d_1-d_2)}{h_2*d_2-h_1*d_1}$$

yuqoridagi F ni topish formulasiga ko'ra

$$F=\frac{h_2+\frac{h_1*h_2(d_1-d_2)}{h_2*d_2-h_1*d_1}}{h_2*d_2}=\frac{h_1-h_2}{h_2*d_2-h_1*d_1}$$

Javob: $F=\frac{h_1-h_2}{h_2*d_2-h_1*d_1}$

@olimpdep

Fan olimpiadalari bo'yicha
iqtidorli o'quvchilar bilan ishlash
DEPARTAMENTI